

Towel Folding SOPs (Medium)

Medium Towels SOP (5x)

Setup

Complete both checklists below before starting any episode.

Hardware checklist

1. Cameras are on and recording.
2. Env camera frame includes the front and back edges of the table and is centered on the table midpoint.
3. Both arms are at the home position with grippers open.
4. Table surface is clear of any objects other than towels.

Materials checklist

1. Five medium towels (16 × 28 inches) are crumpled and placed on the left side of the table at mid depth.
2. Center of the table is clear and used as the working area for straightening and folding.
3. Right side of the table is clear and used as the designated pile location for folded towels.

Workspace layout

Left side: crumpled towel pile input

Center: working area straighten and fold

Right side: folded towel pile output

Steps

Step 1: Move a towel to the working area

Grab one towel from the left pile with the left gripper and move it to the center of the table.

Step 2: Straighten the towel

Goal: towel lies flat on the table with the cross hems along the top and bottom edges.

2.1 Find a corner

With the left gripper, grasp a visible towel corner.

If no corner is visible, grasp any part of the towel and fling it to reorient.

Repeat until a corner is reachable, then grasp it.

2.2 Find the adjacent corner

While the left gripper holds the corner, lift it up.

With the right gripper, grasp the next corner along the same hem.

Make sure to do a proper trace of the side.

2.3 Tension the towel

Bring both grippers to the same height and pull outward in opposite directions to create tension along the gripped hem.

2.4 Fling the towel flat

Fling so that during the drape and fling phase, move the arms in whatever direction feels easiest to get the towel to land flat.

The goal is for the towel to land flat with cross hems along the top and bottom edges.

Step 3: First fold (lengthwise in half)

Definition: crease runs parallel to the longest side, bringing the two long edges together, halving the width.

Left gripper pinch the top left corner of the towel.

Right gripper pinch the top right corner of the towel.

Lift both corners and bring the entire top long edge down to meet the bottom long edge, keeping both grippers at the same height so the fold is even.

Press down gently to set the fold and release both grippers.

Check the fold: the top long edge should be stacked on top of the bottom long edge, with all four corners aligned within 3 cm of each other. If misaligned, lift the top layer and refold.

Expected size after Step 3: ~8 × 28 inches (long narrow strip, horizontal).

Step 4: Second fold (widthwise in half)

Definition: crease runs parallel to the shortest side, bringing the two short edges together, halving the length.

Right gripper pinch the folded strip in the middle of the cloth on the far right side.

Left gripper press down gently on the left side of the cloth stack to stabilize it with minimal force (just enough force to prevent shifting).

Fold the right side over to meet the left side, creating the widthwise fold.

Press down gently to set the fold and release both grippers.

Check the fold: all four corners should be stacked and aligned within 3 cm of each other. If misaligned, lift the top layer and refold.

Expected size after Step 4: ~8 × 14 inches.

Step 5: Stack the folded towel onto the pile

5.1 Check the pile state

If the right-hand pile is empty (first folded towel of the session), no stacking needed. Skip to Step 6.

If the right-hand pile already contains folded towels, continue to Step 5.2.

5.2 Bring the existing pile next to the newly folded towel

With the left gripper, pinch the folded towel on its left side in the center and drag it down and toward the base of the left follower arm and then release.

With the right gripper, pinch the existing pile on its left side in the center and drag it to the center of the table so that the bottom left corner of the pile is closely aligned with the top right corner of the newly folded towel and then release.

5.3 Stack the folded towel onto the pile

Left gripper: pinch the top left corner of the folded towel.

Right gripper: pinch the bottom right corner.

Lift the towel in a U shape (middle sags between the grippers) and place it onto the existing pile, aligning all four corners with the towel below within 3 cm tolerance.

Release both grippers.

Step 6: Move the folded towel or stack to the right side of the table

With the right gripper, pinch the folded towel or stack along the middle of its right edge.

Drag it horizontally to the right until it reaches the designated pile location.

Release the gripper.

Step 7: Return to home and end the episode

If this is not the 5th towel of the session: keep both arms out (do not return to home) and immediately continue to Step 1 with the next towel for continuous folding.

If this is the 5th towel of the session: move both arms back to the home position with grippers open, then end data collection.

After each episode: repeat or reset (session-level workflow)

Check the towel count

Fewer than 5 towels folded this session: return to Step 1 with the next crumpled towel from the left pile.

All 5 towels folded and stacked on the right: reset the workspace before the next session.

Reset the workspace

With recording off:

Manually move all 5 folded towels from the right-hand pile back to the left side of the table.

Flatten and stretch each towel fully (no kinks or folds) to prevent creases over time, then crumple individually so the left side once again contains 5 distinct crumpled towels matching the initial setup state.

Reverify setup

Run through the Setup checklist again before starting the next round of 5 episodes.

SOP Rubric (Steps-only violations)

Use this rubric to score whether each episode followed the SOP steps. These are process/step adherence checks (not quality-of-fold outcome checks), and each item is a violation if it occurs at any point during the episode.

01: Correct pickup source & placement

The towel should be picked from the left input pile and placed in the center working area (not started from center/right, not placed outside the working area).

02: Corner acquisition & reorientation followed

If no corner is visible, the gripper should grasp any part of the towel and fling to reorient, repeating until a corner is reachable, then grasp the corner (no skipping directly to folding while still crumpled).

03: Adjacent corner found on same hem (proper trace)

While holding the first corner, the other gripper should grasp the adjacent corner along the same hem, with a proper trace of the side (no grabbing a non-adjacent corner or random point).

04: Tension step performed before flattening fling

Both grippers should be brought to the same height and pulled outward in opposite directions to create tension along the gripped hem before the flattening fling (no direct fling without tension).

05: Flattening goal orientation met (cross hems top/bottom)

After straightening, the towel should land flat with cross hems along the top and bottom edges (no proceeding to folds while towel is twisted/partially crumpled or hems are left/right).

06: Lengthwise fold performed first (long edge to long edge)

The first fold must be lengthwise in half by lifting top-left and top-right corners and bringing the top long edge to meet the bottom long edge (no widthwise-first, no mid-cloth fold).

07: Lengthwise fold check enforced (≤ 3 cm alignment)

After Step 3, corners should be checked for alignment within 3 cm; if misaligned, the top layer should be lifted and refolded before proceeding (no proceeding without check/refold when visibly off).

08: Widthwise fold mechanics followed (right side over to left)

For Step 4, the right gripper should pinch in the center of the folded far right side, the left gripper stabilizes, then the right side folds over to meet the left (no left-over-right, no folding without stabilization).

09: Widthwise fold check enforced (≤ 3 cm alignment)

After Step 4, corners should be checked for alignment within 3 cm; if misaligned, lift the top layer and refold (no proceeding while misaligned).

10: Conditional stacking logic followed

If the right pile is empty, stacking steps are skipped and the towel proceeds to Step 6. If not empty, stacking steps (5.2–5.3) are performed (no stacking attempt onto empty space; no skipping stacking onto an existing pile).

11: U-shape lift & placement used for stacking

When stacking onto an existing pile, the towel should be lifted by top-left and bottom-right corners into a U shape and placed onto the pile, aligning corners within 3 cm (no flat drag-drop directly on top).

12: Move to designated right-side pile location using right gripper

The folded towel/stack should be pinched along the middle of the right edge with the right gripper and dragged horizontally to the designated right-side pile location (no moving with left gripper; no leaving in center).

13: End-of-episode behavior correct (continuous vs 5th towel)

If not the 5th towel: arms remain out and immediately continue to Step 1. If 5th towel: both arms return to home with grippers open, then end data collection (no returning home early; no failing to return home on 5th).

Annotation subtasks (from SOP)

1. Move one towel from left pile to center
2. Straighten towel flat cross hems top bottom
3. Fold lengthwise in half long edge to long edge
4. Fold widthwise in half short edge to short edge
5. Stack folded towel onto existing pile if any
6. Drag folded towel or stack to right side pile spot
7. Return both arms home end episode